

BANJUL YUNDUM, Gambia

WMO: 617010

Lat: 13.338N

Lon: 16.652W

Elev: 29

StdP: 100.98

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.5	17.2	-0.1	3.7	27.1	1.5	4.2	26.9	10.0	25.6	9.0	25.4	2.5	90	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	8.5	37.9	20.3	36.1	20.4	35.0	20.9	27.7	31.5	27.3	31.0	27.0	30.6	4.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.6	30.2	26.2	21.7	29.5	26.0	21.5	29.3	88.4	32.0	86.6	31.5	85.1	30.8	32.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.4	7.5	6.6	13.3	41.0	2.7	1.4	11.4	42.0	9.8	42.8	8.2	43.6	6.2	44.6		
			9.2	29.8	1.4	1.0	8.2	30.5	7.4	31.1	6.6	31.7	5.6	32.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	25.0	26.2	26.8	26.6	27.0	28.3	27.7	27.3	27.5	28.4	27.4	25.7
	DBStd	1.84	1.71	1.97	2.27	2.06	1.77	1.20	1.12	1.04	1.07	1.08	1.33	1.57
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6199	466	454	520	498	526	548	549	536	525	569	522	487
	CDD18.3	3158	207	221	262	248	267	298	291	277	275	311	272	229
	CDH23.3	29229	1774	1992	2252	1944	2161	3047	2956	2589	2646	3330	2578	1962
	CDH26.7	12567	896	1098	1249	962	951	1205	979	785	913	1351	1204	974
Wind	WSAvg	3.0	3.4	3.8	3.9	4.0	3.7	3.4	2.6	2.4	2.0	1.9	1.9	2.8
Precipitation	PrecAvg	979	0	1	0	0	1	73	206	338	249	68	0	2
	PrecMax	1640	11	11	0	0	10	268	397	600	800	300	7	49
	PrecMin	416	0	0	0	0	0	0	40	102	10	0	0	0
	PrecStd	294	2	2	0	0	2	65	100	127	135	59	2	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	38.9	40.1	39.8	37.2	34.9	33.2	32.2	33.2	35.1	36.1	36.3
		MCWB	19.6	19.7	20.3	21.4	23.1	25.0	26.0	26.5	27.4	26.6	23.3	21.4
	2%	DB	35.0	36.8	38.0	36.2	34.8	33.8	32.2	31.6	32.2	33.9	35.0	35.0
		MCWB	19.1	18.6	19.5	21.2	23.0	24.8	25.9	26.5	27.0	26.7	23.1	20.9
	5%	DB	33.4	35.1	36.0	34.1	33.0	32.9	31.5	31.0	31.8	33.0	33.9	33.8
		MCWB	18.3	18.1	19.1	21.2	23.0	24.8	25.9	26.5	26.9	26.6	23.2	20.4
	10%	DB	31.9	33.3	33.8	32.0	31.6	32.0	30.9	30.2	31.0	32.1	32.8	32.1
		MCWB	17.9	17.7	19.4	21.1	22.8	24.7	25.9	26.1	26.8	26.5	23.3	19.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	22.4	23.8	24.1	25.8	26.8	27.5	27.6	28.2	28.6	27.2	24.8
		MCDB	30.5	32.8	33.5	33.6	31.1	31.6	31.0	30.8	31.8	33.1	32.2	30.4
	2%	WB	20.9	21.0	22.4	23.0	25.0	26.2	26.7	27.1	27.6	27.8	26.4	23.7
		MCDB	29.6	29.8	31.3	31.7	30.5	30.7	30.4	30.4	31.1	31.9	31.0	28.6
	5%	WB	20.1	20.1	21.5	22.3	24.3	25.8	26.4	26.7	27.2	27.4	25.8	22.7
		MCDB	27.5	28.4	30.0	30.1	29.6	30.1	29.9	29.8	30.6	31.4	29.9	27.7
	10%	WB	19.4	19.4	20.7	21.7	23.7	25.5	26.1	26.3	26.8	27.0	25.0	21.8
		MCDB	26.9	27.3	28.8	28.6	28.8	29.6	29.4	29.2	30.0	30.8	28.9	27.5
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.1	14.6	12.9	10.8	8.5	6.9	6.7	7.5	8.5	11.9	14.0
		MCDBR	16.9	17.7	18.0	15.6	12.6	9.3	7.6	7.4	8.4	9.2	13.6	16.3
		MCWBR	6.9	6.1	5.5	4.7	3.9	3.1	3.2	3.4	4.1	3.9	5.8	6.7
	5% WB	MCDBR	15.2	15.2	15.2	13.4	10.7	8.3	7.1	6.9	7.9	8.5	11.3	14.4
		MCWBR	6.6	5.7	5.3	4.5	3.7	3.2	3.4	3.5	4.2	4.0	5.1	6.1
Clear-Sky Solar Irradiance	taub	0.524	0.568	0.656	0.659	0.694	0.705	0.637	0.571	0.559	0.593	0.535	0.503	
	taud	1.921	1.825	1.659	1.708	1.651	1.703	1.856	2.029	2.042	1.926	1.996	2.023	
	Ebn at Noon	774	759	705	701	664	649	698	755	768	730	760	781	
	Edh at Noon	188	215	258	245	255	239	206	176	174	191	173	166	
All-Sky Solar Radiation	RadAvg	5.39	6.03	6.83	7.16	6.93	6.21	5.36	5.12	5.24	5.60	5.37	5.12	
	RadStd	0.28	0.29	0.22	0.19	0.23	0.20	0.26	0.26	0.27	0.25	0.26	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page